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Centre Number

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Student Number

SCEGGS Darlinghurst

2002

**Higher School Certificate
Trial Examination**

Mathematics

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Time allowed - 3 hours.
- An additional 5 minutes reading time is allowed
- Ensure that your student number is on this paper
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Approved scientific calculators should be used. Mathematical templates and geometrical instruments may be used.
- Begin each question on a new page - write your student number at the top of each page.
- The table of standard integrals is printed on the last page.

Questions 1 - 10

Pages 2 - 12

Total marks (120)

Assessment Weighting (40%)

- Attempt all **TEN** questions
- All questions are of equal value.

-
- Answer the questions on the pad paper provided
 - Show ALL necessary working
-

Question 1 (12 marks)**Marks**

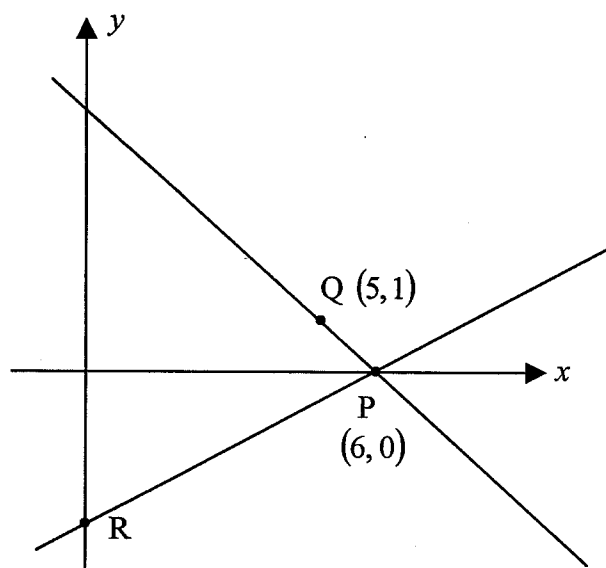
- (a) Write down the exact value of $\cos \frac{\pi}{4}$. 1
- (b) Solve $|5 - 2x| < 9$ and graph the solution on a number line. 2
- (c) Find the primitive function of $\frac{x}{3} + 4$. 2
- (d) Simplify $\frac{3}{x+3} - \frac{1}{x-3}$. 2
- (e) By rationalising the denominator, express $\frac{7}{4-\sqrt{2}}$ in the form $a + b\sqrt{2}$. 3
- (f) Solve the pair of simultaneous equations: 2

$$x - y = 5$$

$$2x + 7y + 8 = 0$$

Question 2 (12 marks)

Marks



The diagram shows the point P (6, 0) and Q (5, 1).

The line PR cuts the y axis at R.

Draw this diagram onto your answer sheet.

- | | | |
|-----|---|---|
| (a) | Find the gradient of PQ. | 1 |
| (b) | RQ is perpendicular to PQ. Find the equation of RQ. | 2 |
| (c) | Find the co-ordinates of R. | 1 |
| (d) | P and R are the endpoints of a diameter of a circle. Find the equation of the circle. | 3 |
| (e) | Show that Q lies on the circle. | 1 |
| (f) | RQTP is a parallelogram. Find the co-ordinates of T. | 1 |
| (g) | Find the area of RQTP. | 3 |

Question 3 (12 marks)

Marks

(a) If α and β are the roots of the equation $2x^2 - 3x - 1 = 0$, find the values of:

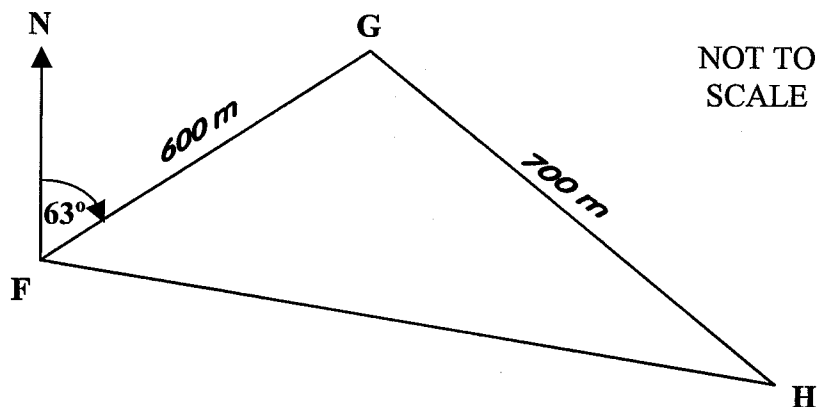
(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ 3

(b) Solve $3 \tan^2 2x = 1$, where $0 \leq x \leq \pi$. 3

(c) Jane is setting up part of an orienteering course.
She follows the course shown in the diagram below.



(i) If $\angle FGH$ is 108° , show that the distance FH to the nearest metre is 1053 metres 1

(ii) Hence, or otherwise, calculate the size of $\angle GFH$ to the nearest degree. 2

(iii) If the bearing of G from F is 063° calculate the bearing of H from F to the nearest degree. 1

Question 4 (12 marks)

Marks

- (a) Evaluate: $\lim_{x \rightarrow 3} \frac{3 + 2x - x^2}{x - 3}$ **2**
- (b) Simplify: $\frac{3^m \times 9^{m+1}}{27^{2m}}$ **2**
- (c) Explain why $x^2 + kx + k - 1 = 0$ has real roots for all values of k . **2**
- (d) Differentiate: $e^x (x + 2)$ **2**
- (e) Evaluate: $\int_{-2}^1 \frac{dx}{\sqrt{2-x}}$ **2**
- (f) Describe the locus of a point moving so that it is equidistant from two fixed points on the number plane. **2**

Question 5 (12 marks)

Marks

- (a) (i) Sketch the function $y = f(x)$ 2

$$\text{where } f(x) = \begin{cases} 1 - x^2 & \text{if } x > 1 \\ x^2 - 1 & \text{if } x \leq 1 \end{cases}$$

- (ii) Find the value of $f(2) + f(1) - f(0)$ 1

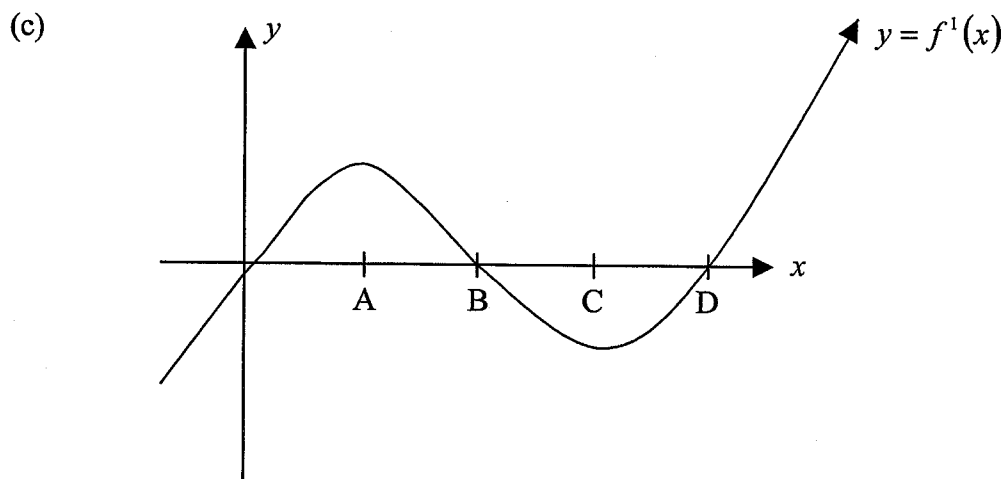
- (b) (i) Show that $\frac{2x+7}{x+3} = 2 + \frac{1}{x+3}$ 1

- (ii) Find the domain and range of $y = \frac{2x+7}{x+3}$ 2

- (iii) Hence sketch $y = \frac{2x+7}{x+3}$ showing all important features. 2

- (iv) Evaluate $\int_0^1 \frac{2x+7}{x+3} dx$ 2

Leave your answer in exact form.

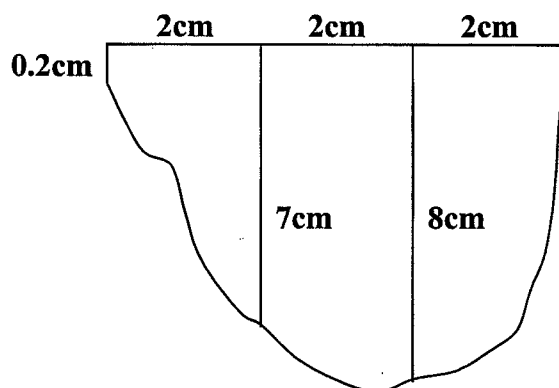


The above diagram shows a sketch of the gradient function of the curve $y = f(x)$. 2
 Draw a sketch of the function $y = f(x)$.

Question 6 (12 marks)

Marks

(a)



The diagram shows the cross-section of a puddle with the depth of the puddle shown in cm at 2cm intervals. The width of the puddle is 6cm.

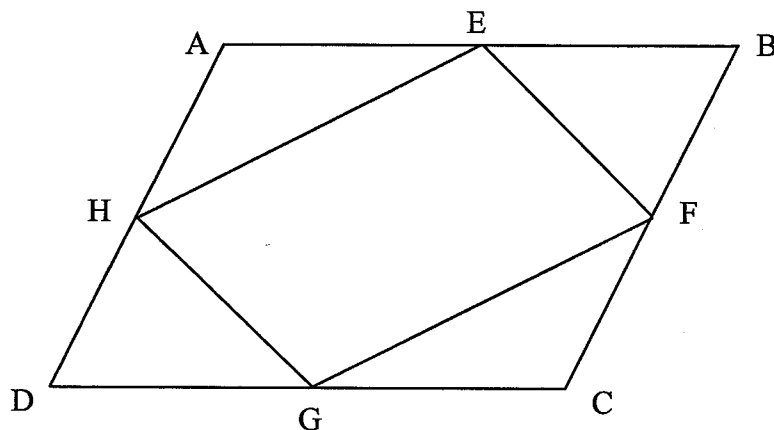
- | | | |
|------|---|----------|
| (i) | Student A uses 3 applications of the trapezoidal rule to find an approximate value for the area of the cross-section. Show the calculations Student A would do. | 2 |
|
 | | |
| (ii) | Student B decides that the cross-section of the puddle approximates the parabolic function $y = x^2 - 6x$. She integrates to find the area of the cross-section between the curve and the x axis. Show the calculations for Student B. | 3 |
|
 | | |
| (b) | In a class of thirty girls, 25 study Mathematics and 20 study History. Each girl studies at least one of these subjects. If a girl is picked at random from this class find the probability she studies: | |
| (i) | both Mathematics and History. | 1 |
|
 | | |
| (ii) | History but not Mathematics. | 1 |

Question 6 continues on page 8

Question 6 (continued)

Marks

- (c) In the diagram below ABCD is a parallelogram. Point E, F, G, H are the midpoints of AB, BC, CD and DA respectively.



- (i) Copy diagram and mark on it the relevant information.

- (ii) Prove that $\triangle AEH$ is congruent to $\triangle FGC$.

3

- (iii) Hence prove that EFGH is a parallelogram.

2

Question 7 (12 marks)

Marks

- (a) (i) Factorise $u^2 - 6u - 16$ 1

- (ii) Hence, or otherwise, solve for x 2

$$[\log_2 x]^2 - 6[\log_2 x] - 16 = 0$$

- (b) Sally deposits \$1700 into a superannuation fund on January 1st, 2001. She makes further deposits of \$1700 on the first of each month up to and including December 1st, 2010. The fund pays compound interest at a monthly rate of 0.75% . In each of the following questions give your answer to the nearest dollar.

- (i) Form a geometric series and hence determine the total amount in the fund on December 31st, 2010. 2

- (ii) If each deposit was increased to \$1800, what difference does it make to the total amount in the fund on December 31st, 2010. 2

- (c) The common ratio r of a geometric progression satisfies the quadratic equation $2r^2 - 3r - 2 = 0$.

- (i) Solve for r : 1

If the sum to infinity of the same progression is 6.

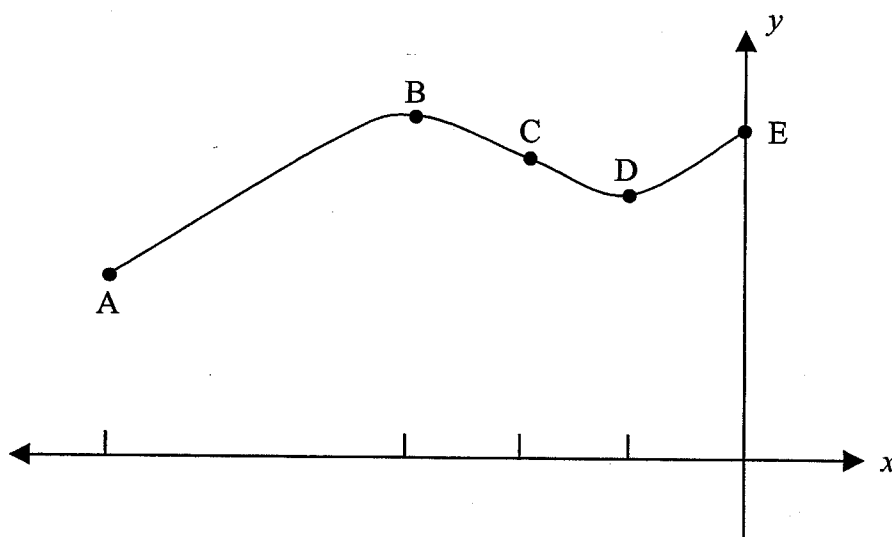
- (ii) Explain why, in this case, r can only take on one value. Hence, state the common ratio r . 2

- (iii) Show that the first term a of this progression, is 9. 2

Question 8 (12 marks)

Marks

- (a) The curve $y = x^3 + 2x^2 + x + 7$ is sketched in the domain $-2 \leq x \leq 0$ with significant points labelled as below.



- | | | |
|-------|--|---|
| (i) | Find points B, C and D and using calculus determine their nature. | 6 |
| (ii) | Between which points is the function decreasing? | 1 |
| (iii) | What is the minimum value of the function? | 1 |
| (iv) | Describe the behaviour of the function for very large positive values of x . | 1 |
- (b) Find the equation of the normal to the curve $y = 3 \ln x$ at the point where $x = 1$. Give your answer in general form. 3

-
- Start a new page
-

Question 9 (12 marks)

Marks

- (a) (i) Sketch the curve $y = x^2 - x - 2$ in the domain $0 \leq x \leq 4$ showing all important points. 2
- (ii) On your diagram for part (i) sketch the line $y = -x + 7$ in the domain $0 \leq x \leq 4$ clearly showing the point of intersection with the curve $y = x^2 - x - 2$ in this domain. 2
- (iii) Find the exact area of the region bounded by the curve, $y = x^2 - x - 2$ the line $y = -x + 7$ and the x axis between $x = 2$ and $x = 4$. 2
- (b) A particle moves on a horizontal line so that its displacement x cm to the right of the origin at time t seconds is $x = t \sin t$.
- (i) Find expressions for the velocity and acceleration of the particle. 2
- (ii) Find the exact velocity of the particle at time $t = \frac{\pi}{4}$. 1
- (iii) What effect does the acceleration have on the velocity of the particle at $t = \frac{\pi}{4}$. 1
- (iv) After the particle leaves the origin is the particle ever at rest? 2
Give reasons for your answer.

Question 10 (12 marks)

Marks

- (a) A bushfire is spreading through a forest and the amount of area burnt since the fire was reported is modelled by the exponential function

$$A = 2e^{kt}$$

where A is the area in hectares and t is the time in hours since the fire was reported. The amount burnt in the first three hours was double the amount burnt since the fire was reported.

- | | | |
|-------|---|---|
| (i) | Find the amount of area burnt when the fire was reported? | 1 |
| (ii) | Find the exact value of k . | 2 |
| (iii) | If the fire was reported at 5am, at what time, to the nearest minute, will 15 hectares have been burnt? | 2 |
| (iv) | Explain why this model is unrealistic. | 2 |
- (b) The rate at which babies are being born has been decreasing at an increasing rate for the last ten years. Draw a graph showing this information. 2
- (c) Find the volume of the solid of revolution formed when the curve $y = \ln 2x$ is rotated about the y axis between $y = 0$ and $y = 2$. Give your answer in exact form. 3

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

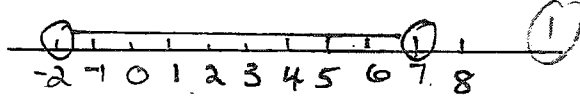
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2002 H.A.C. Trial Mathematics.

Question 1

a) $\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$ (1)
 b) $-9 < 5 - 2x < 9$
 $-14 < -2x < 4$
 $-14 < -2x$ $-2x < 4$
 $-2x > -14$ $x > -2$
 $x < 7$
 $-2 < x < 7$ (1)



c) $\int \frac{x}{3} + 4 \, dx$
 $= \frac{x^2}{6} + 4x + C$ (1)

d) $\frac{3x - 9 - x - 3}{x^2 - 9} = \frac{2x - 12}{x^2 - 9}$ (1)

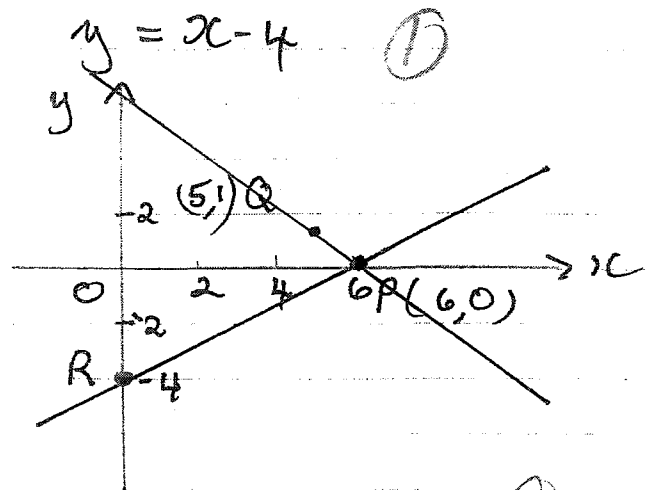
e) $\frac{7}{4 - \sqrt{2}} \times \frac{4 + \sqrt{2}}{4 + \sqrt{2}} = \frac{28 + 7\sqrt{2}}{16 - 2}$
 $= \frac{28 + 7\sqrt{2}}{14} = \frac{4 + \sqrt{2}}{2}$
 $= 2 + \frac{1}{2}\sqrt{2}$ (1)

f) $x - y = 5$ (1)
 $2x + 7y = -8$ (2)
 $\textcircled{1} \times 7$ $7x - 7y = 35$ (3)
 $\textcircled{2} + \textcircled{3}$ $9x = 27$ (1)
 $x = 3$
 sub in $\textcircled{1}$ $3 - y = 5$
 $y = -2$ (1)

Question 2

a) $m_{PQ} = \frac{1}{5-6} = -1$ (1)

b) $m_{RQ} = 1$ (1)
 $y - 1 = 1(x - 5)$



c) $(0, -4) = R$ (1)

d) centre = $(3, -2)$ (1)
 $r = \sqrt{3^2 + (-2+4)^2}$
 $= \sqrt{13}$ (1)

$(x-3)^2 + (y+2)^2 = 13$ (1)

e) $(5-3)^2 + (1+2)^2 = 13$
 $4 + 9 = 13 \checkmark$ (1)

$\therefore Q$ lies on the circle.

f) mid pt. of $PQ = (5\frac{1}{2}, \frac{1}{2})$

Let $T = (x, y_1)$

$\therefore \frac{x_1 + 0}{2} = \frac{11}{2}, \frac{y_1 + (-4)}{2} = \frac{1}{2}$

$x_1 = 11, y_1 = 5$

$\therefore T = (11, 5)$ (1)

g) eqn of RP

$\frac{y+4}{x-0} = \frac{0+4}{6-0} = \frac{2}{3}$

$3y + 12 = 2x$

$2x - 3y - 12 = 0$ (1)

1 dist. of Q to RP

$d = \frac{|10 - 3 - 12|}{\sqrt{4 + 9}}$

$= \frac{5}{\sqrt{13}}$ (1)

$\therefore$ Area $RQTP$

$= 2\sqrt{13} \times \frac{5}{\sqrt{13}} = 10 \text{ m}^2$ (1)

Question 3

a) i) $\alpha + \beta = \frac{3}{2}$ ①

ii) $\alpha\beta = -\frac{1}{2}$ ①

iii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$ ①

$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$ ①

$= \frac{\frac{9}{4} + 1}{-\frac{1}{2}} = -\frac{11}{2}$ ①

b) $\tan 2x = \pm \frac{1}{\sqrt{3}}$ ①

$\therefore 2x = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$ ①

$\therefore x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}$ ①

c) i) $FH^2 = 600^2 + 700^2 -$

$2 \times 600 \times 700 \times \cos 108^\circ$

$\therefore FH \doteq 1053$ ①

ii) $\frac{\sin \angle GFH}{700} = \frac{\sin 108^\circ}{1053}$ ①

$\therefore \angle GFH \doteq 39^\circ$ ①

iii) The bearing of H from F is $102^\circ T$ ①

Question 4

a) $\lim_{x \rightarrow 3} \frac{(3-x)(1+x)}{x-3}$

$= \lim_{x \rightarrow 3} \frac{(3-x)(1+x)}{-(3-x)}$ ①

$= \lim_{x \rightarrow 3} -(1+x)$

$= -4$ ①

b) $\frac{3^m \times 9^{m+1}}{27^{2m}} = \frac{3^m \times 3^{2m+2}}{3^{6m}}$ ①

$= \frac{3^{3m+2}}{3^{6m}} = 3^{-3m+2}$ ①

c) $\Delta = b^2 - 4ac = 4 - 4$

$\Delta = (b-2)^2$ ①

which is a perfect square $\therefore \Delta$ is always +ve or equal to 0 ①

$\therefore$ the quadratic eqn. has real roots for all values of k .

d) Let $y = e^{x(x+2)}$
 $\frac{dy}{dx} = e^{x(x+2)} + e^{x(x+2)} \cdot x$ ①

$= e^x (x+3)$ ①

e) $\int \frac{dx}{\sqrt{2-x}} = \left[-2(2-x)^{\frac{1}{2}} \right]_{-2}^1$ ①

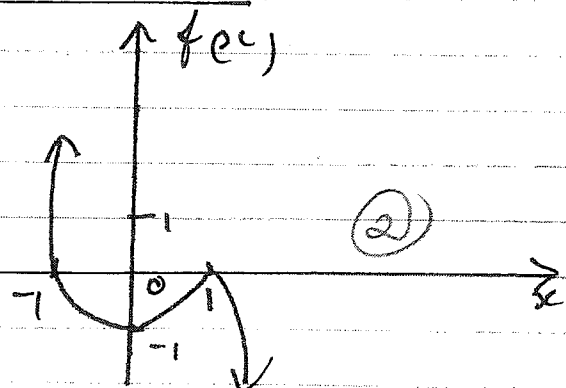
$= (-2) - (-2\sqrt{3})$

$= -2 + 2\sqrt{3}$ ①

f) The locus is the perpendicular bisector of the interval joining the 2 fixed points ②

Question 5

a) i)

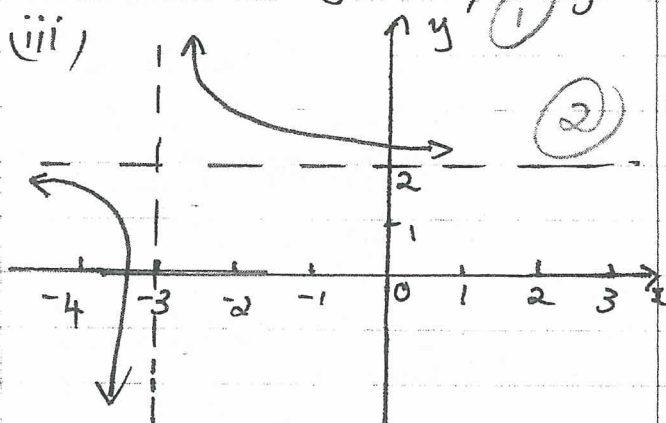


ii) $f(2) + f(1) - f(0)$
 $= 1 - 4 + 1 = -2$ ①

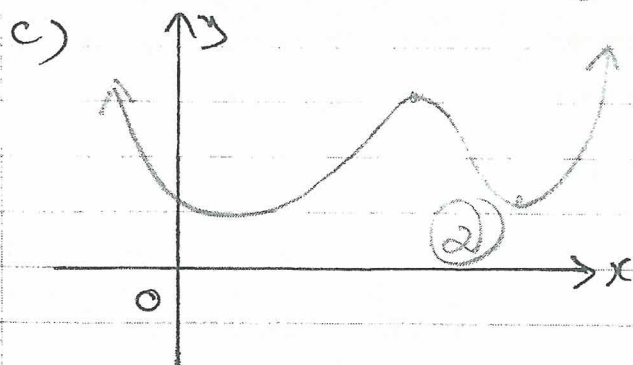
$$\begin{aligned}
 \text{b) i) L.H.S.} &= \frac{2x+7}{x+3} \\
 &= \frac{2x+6+1}{x+3} \\
 &= \frac{2(x+3)+1}{x+3} \quad (1) \\
 &= 2 + \frac{1}{x+3} = \text{R.H.S.}
 \end{aligned}$$

ii) D: all real x except $x = -3$

R: all real y except $y = 2$



$$\begin{aligned}
 \text{iv) } \int_0^1 2 + \frac{1}{x+3} dx & \\
 &= [2x + \ln(x+3)]_0^1 \\
 &= 2 + \ln 4 - \ln 3 \quad (1)
 \end{aligned}$$



Question 6

a) i) $h = 2$

x	$f(x)$	w	$w \times f(x)$
0	0.2	1	0.2
2	7	2	14
4	8	2	16
6	0	1	0

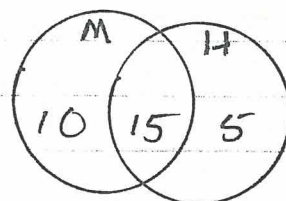
$$30.2$$

$$\text{Area} \doteq \frac{\pi}{2} \times 30.2 = 30.2 \text{ cm}^2$$

$$\begin{aligned}
 \text{ii) } A &= \left| \int_0^6 x^2 - 6x dx \right| \quad (1) \\
 &= \left| \left[\frac{x^3}{3} - 3x^2 \right]_0^6 \right| \quad (1)
 \end{aligned}$$

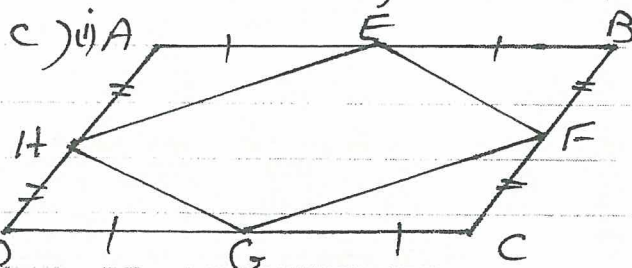
$$= |72 - 36| = |36| = 36$$

b)



$$\text{i) } P(\text{both}) = \frac{15}{30} = \frac{1}{2} \quad (1)$$

$$\text{ii) } P(H \text{ not } M) = \frac{5}{30} = \frac{1}{6} \quad (1)$$



ii) In Δ 's AEH and FGC
 $AB = DC$ (opp. sides of $\parallel$ gram)
 $\therefore AE = GC$ (E & G mid pts. of opposite sides of $\parallel$ gram)
 $AD = BC$ (opp. sides of $\parallel$ gram)
 $\therefore AH = FC$ (H & F mid pts. of opp. sides of $\parallel$ gram)
 $\angle HAE = \angle FGC$ (opp $\angle$'s of $\parallel$ gram)
 $\therefore \Delta AEH \equiv \Delta FGC$ (SAS)

(3)

iii) Similarly it can be proved $\triangle HOG \cong \triangle EBF$
 $\therefore HE = GF$ (matching sides of $\cong \Delta$'s) and
 $HG = FE$ (matching sides of $\cong \Delta$'s) (2)
 $\therefore EFGH$ is a parallelogram

Question 7

- a) i) $(u-8)(u+2)$ (1)
 ii) Let $\log_2 x = u$

$$u^2 - 6u - 16 = 0$$

$$(u-8)(u+2) = 0$$

$$\therefore u = 8, -2$$

$$\log_2 x = 8$$

$$\therefore x = 2^8 = 256$$
 (1)
$$\log_2 x = -2$$

$$\therefore x = 2^{-2} = \frac{1}{4}$$
 (1)

b) i) $A_1 = \$17000 \times 1.0075^{120}$
 $A_2 = \$17000 \times 1.0075^{119}$
 and so on

$$A_{120} = \$17000 \times 1.0075^{120}$$

... at the end of 10 years the investment is worth

$$\$1700(1.0075^{120} + 1.0075^{119} + \dots + 1.0075)$$

$$= 1700 \times 1.0075 \frac{(1.0075^{120} - 1)}{1.0075 - 1}$$

$$= 1700 \times 194.96563$$

$$\therefore \text{total amount} = \$331441.58$$
 (1)

ii) New amount = $\$1800 \times 194.96563$

$$= \$350938.14$$
 (1)

$$\therefore \text{Diff} = \$19496.56$$
 (1)
 extra

ci) $(2r+1)(r-2)=0$
 $\therefore r = -\frac{1}{2}, 2$ (1)

ii) For the series to have a S_{∞} $-1 < r < 1$ (1)

$$\therefore r = -\frac{1}{2}$$
 (1)

iii) $b = \frac{a}{1 - (-\frac{1}{2})}$ (1)

$$6 = \frac{a}{\frac{3}{2}}$$
 (1)

$$\therefore a = 6 \times \frac{3}{2} = 9$$

Question 8

a) i) $\frac{dy}{dx} = 3x^2 + 4x + 1$

$$3x^2 + 4x + 1 = 0$$
 (1)

$$(3x+1)(x+1) = 0$$

$$\therefore x = -\frac{1}{3}, -1$$

$$y = 6\frac{23}{27}, 7$$

$$\frac{d^2y}{dx^2} = 6x + 4$$
 (1)

when $x = -1$, $= -ve$

$$\therefore \text{max. at } (-1, 7)$$
 (1) B

when $x = -\frac{1}{3}$, $= +ve$

$$\therefore \text{min at } (-\frac{1}{3}, 6\frac{23}{27})$$
 (1) D

$$6x + 4 = 0$$

$$\therefore x = -\frac{2}{3}$$

x	-1	$-\frac{2}{3}$	$-\frac{1}{3}$
$\frac{d^2y}{dx^2}$	$-ve$	0	$+ve$

$$\therefore \text{pt inflex } (-\frac{2}{3}, 6\frac{25}{27})$$
 (1) C

ii) B and D

 (1)

iii) when $x = -2$

$$y = (-2)^3 + 2x(-2)^2 - 2$$

$$\therefore \text{min value is } 5$$
 (1)

(iv) as $x \rightarrow \infty$
 $y \rightarrow \infty$

b) $y = 3 \ln x$
 $\frac{dy}{dx} = \frac{3}{x}$

when $x = 1$
 $m_{\tan} = 3 \therefore m_{\text{norm}} = -\frac{1}{3}$

$y - 0 = -\frac{1}{3}(x - 1)$
 $3y = -x + 1$
 $x + 3y - 1 = 0$

Question 9

a) i) $y = x^2 - x - 2$

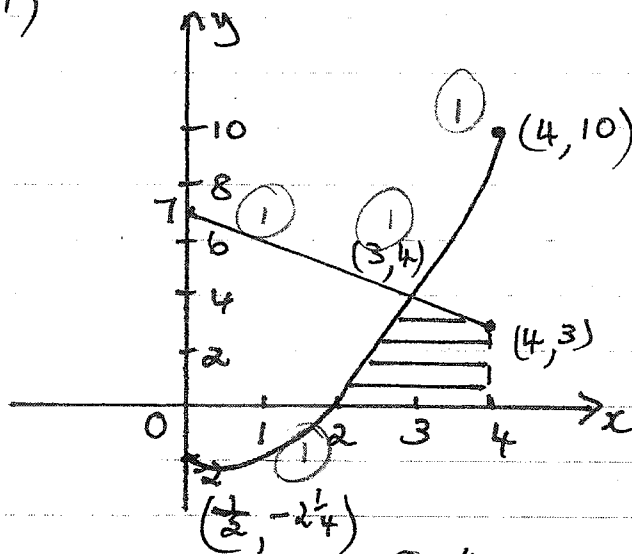
$x^2 - x - 2 = 0$

$(x - 2)(x + 1) = 0$

$\therefore x = 2, -1$

$(4, 10)$

(ii)



(iii) $A = \int_2^4 x^2 - x - 2 dx + \int_2^4 -x + 7 dx$
 $= \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_2^4 + \left[-\frac{x^2}{2} + 7x \right]_2^4$
 $= (9 - \frac{9}{2} - 6) - (\frac{8}{3} - 2 - 4) + (-8 + 28) - (-2 + 21)$
 $= 5\frac{1}{3} \text{ m}^2$

b) i) $x = t \sin t$
 $\dot{x} = \sin t + t \cos t$

$\ddot{x} = \cos t + \cos t - t \sin t$

$\ddot{x} = 2 \cos t - t \sin t$

(ii) when $t = \frac{\pi}{4}$

$\dot{x} = \frac{1}{\sqrt{2}} + \frac{\pi}{4\sqrt{2}} \text{ cm/sec}$

(iii) when $t = \frac{\pi}{4}$

$\ddot{x} = \frac{1}{\sqrt{2}} \times 2 - \frac{\pi}{4} \times \frac{1}{\sqrt{2}}$

$= \frac{2}{\sqrt{2}} - \frac{\pi}{4\sqrt{2}} \text{ cm/s}^2 = +ve$

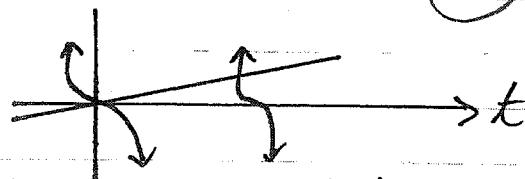
$\therefore$ acceleration is

speeding up the particle

(iv) $0 = t \cos t + \sin t$

$-\sin t = t \cos t$

$-\tan t = t$



Yes from sketch the particle is at rest again after the origin

Question 10

a) i) $A = 2e^{kt}$

$t = 0, A = 2 \text{ hr}$

(ii) $4 = 2e^{3h}$

$2 = e^{3h}$

$\ln 2 = 3h$

$h = \frac{1}{3} \ln 2$

(iii) $15 = 2e^{\frac{1}{3} \ln 2 t}$

$\ln \frac{15}{2} = \frac{1}{3} \ln 2 t$

$t = \frac{\ln \frac{15}{2}}{\frac{1}{3} \ln 2}$

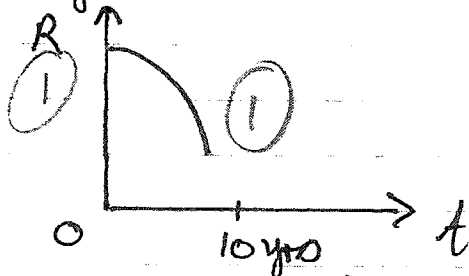
$= 8.7 \text{ hrs} = 8 \text{ hrs } 43 \text{ min}$

$\therefore 1.43 \text{ p.m.}$

(iv) this model assumes the bushfire's will continue burning at the same

② rate under the same conditions for ever which does not happen due to weather and different type of vegetation

b)



decreasing $y' < 0$
increasing rate $y'' > 0$

c)

$$y = \ln 2x$$

$$2x = e^y$$

$$x = \frac{1}{2} e^y$$

$$x^2 = \frac{1}{4} e^{2y}$$

$$V = \pi \int_0^2 \frac{1}{4} e^{2y} dy$$

$$= \pi \left[\frac{1}{8} e^{2y} \right]_0^2$$

$$= \pi \left(\frac{1}{8} e^4 - \frac{1}{8} \right) = \frac{\pi}{8} (e^4 - 1)$$

cu units